

Laboratorio 7

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void function(int n) {
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    int i, j, k, counter = 0;
```

```
    for (i = n/2; i <= n; i++) {
```

```
        for (j = 1; j+n/2 <= n; j++) {
```

```
            for (k = 1; k <= n; k = k*2) {
```

```
                counter++;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

1

$\frac{n}{2} + 2$

$\sum_{i=n/2}^n (\frac{n}{2} + 1) = \frac{n^2}{4} + n + 1$

$\sum_{i=n/2}^n \sum_{j=i}^{n/2} (\log_2 n + 2) = (\frac{n^2}{4} + \frac{n}{2})(\log_2 n + 2)$

$\sum_{i=n/2}^n \sum_{j=i}^{n/2} (\log_2 n + 1) = (\frac{n^2}{4} + \frac{n}{2})(\log_2 n + 1)$

$$T(n) = 1 + (\frac{n}{2} + 2) + (\frac{n^2}{4} + n + 1) + (\frac{n^2}{4} + \frac{n}{2})(\log_2 n + 2) + (\frac{n^2}{4} + \frac{n}{2})(\log_2 n + 1)$$

$$= \frac{n^2}{4} + \frac{3n}{2} + 4 + (\frac{n^2}{4} + \frac{n}{2})(2\log_2 n + 3)$$

$$= (\frac{n^2}{2} + n)\log_2 n + n^2 + 3n + 4 \quad g(n) = n^2 \log_2 n$$

$$C_1 \cdot n^2 \log_2 n \leq T(n) \leq C_2 \cdot n^2 \log_2 n$$

$$= \left(\frac{n^2}{2}\right) \log_2 n + n \log_2 n + n^2 + 3n + 4$$

$\frac{1}{2} + 1 + 1 + 3 + 4 = 9.5$

$$C_1 = \frac{1}{2} \quad C_2 = 9.5 \quad n_0 = 2$$

$$\forall n \geq 2, \frac{n^2}{2} \log_2 n \leq T(n) \leq 9.5n^2 \log_2 n$$

$$T(n) \in \Theta(n^2 \log n)$$

```

void function(int n) {
    if (n <= 1) return;
    int i, j;
    for (i = 1; i <= n; i++) {  $\rightarrow n+1$ 
        for (j = 1; j <= n; j++) {  $\rightarrow n$ 
            printf("Sequence\n");  $\rightarrow n$ 
            break;  $\rightarrow n$ 
        }
    }
}

```

Best case

$$T(n) = 1$$

worst case

$$T(n) = 1 + (n+1) + n + n + n$$

$$T(n) = 4n + 2$$

$$g(n) = n$$

$$\forall n \geq 2$$

$$4n \leq T(n) \leq 6n$$

$$T(n) \in \Theta(n)$$

$$C_1 = 4 \quad C_2 = 6 \quad n_0 = 2$$

```

void function(int n) {
    int i, j;
    for (i = 1; i <= n/3; i++) {  $\rightarrow \frac{n}{3} + 1$ 
        for (j = 1; j <= n; j += 4) {  $\rightarrow \sum_{i=1}^{\frac{n}{3}} (\frac{n}{4} + 1) = \frac{n^2}{12} + \frac{n}{3}$ 
            printf("Sequence\n");
        }
    }
}

```

$\rightarrow \sum_{i=1}^{\frac{n}{3}} \frac{n}{4} = \frac{n}{3} + \frac{n}{4}$

$$T(n) = \left(\frac{n}{3} + 1\right) + \left(\frac{n^2}{12} + \frac{n}{3}\right) + \frac{n^2}{12} = \frac{n^2}{6} + \frac{2n}{3} + 1 \quad g(n) = n^2$$

$$C_1 = \frac{1}{6} \quad C_2 = \frac{11}{6} \quad n_0 = 1$$

$$\forall n \geq 1, \quad \frac{1}{6}n^2 \leq T(n) \leq \frac{11}{6}n^2$$

$$T(n) \in \Theta(n^2)$$